

1.3

9. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.

a) $p \rightarrow \neg q$

11. Show that each of these conditional statements is a tautology by using truth tables.

a) $(p \wedge q) \rightarrow p$

b) $p \rightarrow (p \vee q)$

1.4

9. Let $P(x)$ be the statement “ x can speak Russian” and let $Q(x)$ be the statement “ x knows the computer language C++.” Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.
- a) There is a student at your school who can speak Russian and who knows C++.
 - b) There is a student at your school who can speak Russian but who doesn’t know C++.
 - c) Every student at your school either can speak Russian or knows C++.
 - d) No student at your school can speak Russian or knows C++.

1.6

- 9.** For each of these collections of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.
- a)** “If I take the day off, it either rains or snows.” “I took Tuesday off or I took Thursday off.” “It was sunny on Tuesday.” “It did not snow on Thursday.”
- b)** “If I eat spicy foods, then I have strange dreams.” “I have strange dreams if there is thunder while I sleep.” “I did not have strange dreams.”

13. For each of these arguments, explain which rules of inference are used for each step.

- a) “Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job.”
- b) “Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution.”

2.1

29. Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find

a) $A \times B$.

b) $B \times A$.

2.2

25. Prove the first distributive law from Table 1 by showing that if A , B , and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

2.6

27. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Find

a) $\mathbf{A} \vee \mathbf{B}.$

b) $\mathbf{A} \wedge \mathbf{B}.$

c) $\mathbf{A} \odot \mathbf{B}.$

3.3

3.1

25. Describe an algorithm that will count the number of 1s in a bit string by examining each bit of the string to determine whether it is a 1 bit.

25. This algorithm is straightforward.

```
procedure count ones( $a_1a_2 \dots a_n$  : bit string)
  count := 0 {no 1's yet}
  for  $i := 1$  to  $n$ 
    if  $a_i = 1$  then count := count + 1
  return count { count contains the number of 1's }
```

9. Give a big- O estimate for the number of comparisons used by the algorithm that determines the number of 1s in a bit string by examining each bit of the string to determine whether it is a 1 bit (see Exercise 25 of Section 3.1).

27. The **ternary search algorithm** locates an element in a list of increasing integers by successively splitting the list into three sublists of equal (or as close to equal as possible) size, and restricting the search to the appropriate piece. Specify the steps of this algorithm.

- 27.** We start with the pseudocode for binary search given in the text and modify it. In particular, we need to compute two middle subscripts (one third of the way through the list and two thirds of the way through) and compare x with two elements in the list. Furthermore, we need special handling of the case when there are two elements left to be considered. The following pseudocode is reasonably straightforward.

```

procedure ternary search( $x$  : integer,  $a_1, a_2, \dots, a_n$  : increasing integers)
 $i := 1$ 
 $j := n$ 
while  $i < j - 1$ 
     $l := \lfloor (i + j)/3 \rfloor$ 
     $u := \lfloor 2(i + j)/3 \rfloor$ 
    if  $x > a_u$  then  $i := u + 1$ 
    else if  $x > a_l$  then
         $i := l + 1$ 
         $j := u$ 
    else  $j := l$ 
if  $x = a_i$  then  $location := i$ 
else if  $x = a_j$  then  $location := j$ 
else  $location := 0$ 
return  $location$ 
{  $location$  is the subscript of the term equal to  $x$  (0 if not found) }

```

3.3

25. Describe the worst-case time complexity, measured in terms of comparisons, of the ternary search algorithm described in Exercise 27 of Section 3.1.

- 25.** We will count comparisons of elements in the list to x . (This ignores comparisons of subscripts, but since we are only interested in a big- O analysis, no harm is done.) Furthermore, we will assume that the number of elements in the list is a power of 3, say $n = 3^k$. Just as in the case of binary search, we need to determine the maximum number of times the **while** loop is iterated. Each pass through the loop cuts the number of elements still being considered (those whose subscripts are from i to j) by a factor of 3. Therefore after k iterations, the active portion of the list will have length 1; that is, we will have $i = j$. The loop terminates at this point. Now each iteration of the loop requires two comparisons in the worst case (one with a_u and one with a_l). Two more comparisons are needed at the end. Therefore the number of comparisons is $2k + 2$, which is $O(k)$. But $k = \log_3 n$, which is $O(\log n)$ since logarithms to different bases differ only by multiplicative constants, so the time complexity of this algorithm (in all cases, not just the worst case) is $O(\log n)$.

4.1

29. Find $a \mathbf{div} m$ and $a \mathbf{mod} m$ when

a) $a = 228, m = 119.$

31. Find the integer a such that

a) $a \equiv -15 \pmod{27}$ and $-26 \leq a \leq 0.$

4.2

1. Convert the decimal expansion of each of these integers to a binary expansion.

a) 231

17. Convert $(7345321)_8$ to its binary expansion and $(10\ 1011\ 1011)_2$ to its octal expansion.

4.3

33. Use the Euclidean algorithm to find

b) $\gcd(111, 201)$.

4.4

11. Solve each of these congruences using the modular inverses found in parts (b), (c), and (d) of Exercise 5.

a) $19x \equiv 4 \pmod{141}$

b) $55x \equiv 34 \pmod{89}$

4.6

5. Decrypt these messages encrypted using the shift cipher

$$f(p) = (p + 10) \bmod 26.$$

a) CEBBOXNOB XYG

5.1

21. Prove that $2^n > n^2$ if n is an integer greater than 4.

6.2

- 11.** What is the minimum number of students, each of whom comes from one of the 50 states, who must be enrolled in a university to guarantee that there are at least 100 who come from the same state?

